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On the determination of residual stress in an elastic body

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1. Introduction

Residual stress is the stress supported by a body in a fixed reference configuration in the absence of external forces. * Thus the residual stress $\mathring{\boldsymbol{T}}$ satisfies the equilibrium equations **

$$\left.\begin{aligned} \nabla \cdot \mathring{\boldsymbol{T}} &= \boldsymbol{0}, \\ \mathring{\boldsymbol{T}} &= \mathring{\boldsymbol{T}}^T \end{aligned}\right\} \tag{1.1}$$

on the body B and the zero traction condition

$$\mathring{\boldsymbol{T}}\boldsymbol{n} = \boldsymbol{0} \tag{1.2}$$

on its boundary ∂B, where $\boldsymbol{n}$ is the outward unit normal.

Some insight into the subtleties of residual stress can be gained by calculation of its volume average

$$\boldsymbol{T}_M(B) = \frac{1}{V(B)} \int_B \mathring{\boldsymbol{T}} \, \mathrm{d}V,$$

where $V(B)$ is the volume of B. The mean stress theorem (see, e.g. [2,§18]) states that

$$\mathring{\boldsymbol{T}}_M(B) = \frac{1}{V(B)} \left[\int_{\partial B} (\boldsymbol{p} - \mathring{\boldsymbol{p}}) \otimes (\mathring{\boldsymbol{T}}\boldsymbol{n}) \, \mathrm{d}A_p - \int_B (\boldsymbol{p} - \mathring{\boldsymbol{p}}) \otimes (\nabla \cdot \mathring{\boldsymbol{T}}) \, \mathrm{d}V_p \right]$$

where $\mathring{\boldsymbol{p}}$ is a fixed reference point. Consequently, with (1.1a) and (1.2),

$$\mathring{\boldsymbol{T}}_M(B) = \boldsymbol{0};$$

i.e., *the mean residual stress is always zero.* This implies that the cartesian components of $\mathring{\boldsymbol{T}}$ must take on both positive and negative values in the body. In particular, *a nonzero residual stress field cannot be uniform.*

Our goal is to construct a theoretical framework with which to explore the possibility of the nondestructive mechanical determination of residual stress. The body is supposed to support an unknown residual stress field in its reference configuration, and it is assumed that the body responds in a linearly elastic manner to small deformations from the reference configuration. The residual stress appears explicitly in

* It would be physically realistic to admit the presence of a nonzero body force field, such as gravity, but we do not do so here.

** Throughout we use the notation and terminology of GURTIN's text [1].

the corresponding constitutive equation, and we treat it as constitutive information to be determined by an appropriate combination of theory and experiment. Since a nonzero residual stress field is necessarily nonuniform, the response of the body will always be inhomogeneous. Our approach is completely statical. For dynamic techniques reference may be made to the recent survey of PAO, SACHSE, and FUKUOKA [3].

In Section 2, starting with the theory of finite elasticity, we provide a careful derivation of the linearized constitutive equation *

$$\boldsymbol{S} = \mathring{\boldsymbol{T}} + \boldsymbol{W}\mathring{\boldsymbol{T}} + \tfrac{1}{2}(\boldsymbol{E}\mathring{\boldsymbol{T}} - \mathring{\boldsymbol{T}}\boldsymbol{E}) + \mathsf{L}[\boldsymbol{E}],$$

which is appropriate to small elastic deformations from the residually stressed reference configuration. We make no assumptions concerning how the residual stress arises in the first place; in particular, it need not be the result of prior elastic behavior.

In Section 3 we consider a body that has the shape of a right circular cylinder in its residually stressed reference configuration. In addition we assume that the residual stress field has the form ** $\mathring{\boldsymbol{T}} = \mathring{\boldsymbol{T}}(r, \theta)$ and that the incremental elasticity tensor L has an isotropic and homogeneous structure. Then, roughly speaking, we are able to show that the incremental elasticities are determined by the tension test and that the residual stress field is determined *pointwise* by the torsion test. In the latter case pointwise measurement of the shear stress distribution applied to the end of the cylinder is required. We also obtain lower bounds for the components of the residual stress in terms of the applied torque and the angle of twist. This provides a positive indicator for the presence of residual stress by means of standard measurements.

2. Derivation of the linear theory

In this section we derive the linear approximation to the field equations of finite elasticity under the assumptions that the stress in the reference configuration is not necessarily zero and that the gradient of the displacement from the reference configuration is small.

2.1 Finite elasticity with residual stress

We identify the body with the regular region of space B that it occupies in a fixed reference configuration. It is presumed that in this reference configuration the body is at rest and that no external forces are present. The residual stress $\mathring{\boldsymbol{T}}$ is the Cauchy stress field in the reference configuration. As noted previously, it satisfies (1.1) and (1.2).

A deformation $\boldsymbol{f}$ from the reference configuration is a smooth one-to-one mapping that carries point $\boldsymbol{p} \in B$ into point $\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{p})$. As usual the deformation gradient

$$\boldsymbol{F} = \nabla \boldsymbol{f}$$

is required to meet $\det \boldsymbol{F} > 0$.

* The symbols used here have their customary meanings, see Section 2 for precise definitions.

** Here we use cylindrical coordinates appropriate to the undeformed body. Note that this form includes the technologically important case $\boldsymbol{T} = \boldsymbol{T}(r)$ which can be expected in drawn wire. A few observations concerning other forms for the residual stress are also made.

We assume that the body responds elastically to deformation from the reference configuration * so the constitutive equation for the Cauchy stress is

$$\boldsymbol{T}(\boldsymbol{x}) = \hat{\boldsymbol{T}}(\boldsymbol{F}(\boldsymbol{p}), \boldsymbol{p}). \tag{2.1.1}$$

Since the residual stress $\mathring{\boldsymbol{T}}$ is defined as the stress in the reference configuration,

$$\mathring{\boldsymbol{T}}(\boldsymbol{p}) = \hat{\boldsymbol{T}}(\boldsymbol{1}, \boldsymbol{p}). \tag{2.1.2}$$

In what follows, the explicit dependence of the constitutive relations on $\boldsymbol{p}$ will be suppressed except where its appearance is needed for clarity.

For the elastic response to be independent of the observer, the constitutive function must satisfy

$$\hat{\boldsymbol{T}}(\boldsymbol{QF}) = \boldsymbol{Q}\hat{\boldsymbol{T}}(\boldsymbol{F})\boldsymbol{Q}^T \tag{2.1.3}$$

for all $\boldsymbol{F} \in \mathrm{Lin}^+$ and $\boldsymbol{Q} \in \mathrm{Orth}^+$.

The material symmetry group at $\boldsymbol{p}$, $\mathscr{G}_{\mathrm{p}}$, is the set of all proper orthogonal tensors $\boldsymbol{Q}$ that satisfy

$$\hat{\boldsymbol{T}}(\boldsymbol{FQ}, \boldsymbol{p}) = \hat{\boldsymbol{T}}(\boldsymbol{F}, \boldsymbol{p}) \tag{2.1.4}$$

for all $\boldsymbol{F} \in \mathrm{Lin}^+$. Combining material symmetry and observer independence (2.1.3), we find that the response function $\hat{\boldsymbol{T}}$ is invariant under $\mathscr{G}$; i.e.,

$$\hat{\boldsymbol{T}}(\boldsymbol{QFQ}^T) = \boldsymbol{Q}\hat{\boldsymbol{T}}(\boldsymbol{F})\boldsymbol{Q}^T \tag{2.1.5}$$

for all $\boldsymbol{Q} \in \mathscr{G}_{\mathrm{p}}$ and $\boldsymbol{F} \in \mathrm{Lin}^+$. In view of (2.1.2), evaluation of (2.1.5) at $\boldsymbol{F} = \boldsymbol{1}$ results in

$$\boldsymbol{Q}\mathring{\boldsymbol{T}}(\boldsymbol{p}) = \mathring{\boldsymbol{T}}(\boldsymbol{p})\boldsymbol{Q} \tag{2.1.6}$$

for all $\boldsymbol{Q} \in \mathscr{G}_{\mathrm{p}}$; i.e., the *residual stress at a point commutes with every element of the material symmetry group at that point.* COLEMAN and NOLL [4] have found the forms for residual stress fields for various specific symmetry groups, and HOGER [5] has considered the additional restrictions imposed by the equilibrium and zero traction conditions (1.1) and (1.2).

It will be convenient here to work with the first Piola-Kirchhoff stress $\boldsymbol{S}$ defined by

$$\boldsymbol{S} = (\det \boldsymbol{F})\boldsymbol{T}\boldsymbol{F}^{-T} = (\det \boldsymbol{F})\hat{\boldsymbol{T}}(\boldsymbol{F})\boldsymbol{F}^{-T} \equiv \hat{\boldsymbol{S}}(\boldsymbol{F}). \tag{2.1.7}$$

In what follows the response function $\hat{\boldsymbol{S}}(\cdot, \boldsymbol{p})$ is assumed to be differentiable for each $\boldsymbol{p} \in B$. At $\boldsymbol{F} = \boldsymbol{1}$,

$$\hat{\boldsymbol{S}}(\boldsymbol{1}) = \hat{\boldsymbol{T}}(\boldsymbol{1}) = \mathring{\boldsymbol{T}}. \tag{2.1.8}$$

In terms of $\hat{\boldsymbol{S}}$, observer independence requires that

$$\hat{\boldsymbol{S}}(\boldsymbol{QF}) = \boldsymbol{Q}\hat{\boldsymbol{S}}(\boldsymbol{F}) \tag{2.1.9}$$

for every $\boldsymbol{F} \in \mathrm{Lin}^+$ and $\boldsymbol{Q} \in \mathrm{Orth}^+$; and $\boldsymbol{Q} \in \mathscr{G}$ is equivalent to

$$\hat{\boldsymbol{S}}(\boldsymbol{QFQ}^T) = \boldsymbol{Q}\hat{\boldsymbol{S}}(\boldsymbol{F})\boldsymbol{Q}^T \tag{2.1.10}$$

for all $\boldsymbol{F} \in \mathrm{Lin}^+$.

The local forms of force and moment balance are expressed in terms of $\boldsymbol{S}$ as

$$\nabla \cdot \boldsymbol{S} + \boldsymbol{b} = \boldsymbol{0} \tag{2.1.11}$$

* It is important to understand that we are not assuming that the presence of the residual stress is due to elastic behavior.

and

$$\boldsymbol{S}\boldsymbol{F}^T = \boldsymbol{F}\boldsymbol{S}^T, \tag{2.1.12}$$

respectively, where $\boldsymbol{b}(\boldsymbol{p})$ is the body force density at $\boldsymbol{f}(\boldsymbol{p})$ measured per unit volume in the reference configuration.

If the body is hyperelastic, then there exists a strain-energy function $\hat{\sigma}$:Lin$^+ \times B \to \mathbb{R}$ such that

$$\hat{\boldsymbol{S}}(\boldsymbol{F}, \boldsymbol{p}) = \frac{\partial \hat{\sigma}}{\partial \boldsymbol{F}}(\boldsymbol{F}, \boldsymbol{p}). \tag{2.1.13}$$

Here, as usual, the meaning of the tensor $\dfrac{\partial \hat{\sigma}}{\partial \boldsymbol{F}}$ is given by

$$D_F \hat{\sigma}(\boldsymbol{F}, \boldsymbol{p})[\boldsymbol{A}] = \frac{\partial \hat{\sigma}}{\partial \boldsymbol{F}}(\boldsymbol{F}, \boldsymbol{p}) \cdot \boldsymbol{A},$$

where the inner product on Lin is defined to be

$$\boldsymbol{A} \cdot \boldsymbol{B} = \operatorname{tr}(\boldsymbol{A}^T \boldsymbol{B}).$$

2.2. The Linearized Constitutive Equation

The behavior of the response function $\hat{\boldsymbol{S}}$ near the reference configuration is governed by the **elasticity tensor,** which for each $\boldsymbol{p} \in B$ is the linear transformation $\mathbf{C}(\boldsymbol{p})$: Lin $\to$ Lin defined by

$$\mathbf{C}(\boldsymbol{p})[\boldsymbol{A}] = D_F \hat{\boldsymbol{S}}(\mathbf{1}, \boldsymbol{p})[\boldsymbol{A}]. \tag{2.2.1}$$

The following four properties of the elasticity tensor will be used:

$$\mathbf{C}[\boldsymbol{W}] = \boldsymbol{W}\mathring{\boldsymbol{T}} \qquad \text{for all } \boldsymbol{W} \in \text{Skw}; \tag{2.2.2}$$

$$\text{skw } \mathbf{C}[\boldsymbol{E}] = \tfrac{1}{2}(\boldsymbol{E}\mathring{\boldsymbol{T}} + \mathring{\boldsymbol{T}}\boldsymbol{E}) \qquad \text{for all } \boldsymbol{E} \in \text{Sym}; \tag{2.2.3}$$

$$\mathbf{C}[\boldsymbol{Q}\boldsymbol{H}\boldsymbol{Q}^T] = \boldsymbol{Q}\mathbf{C}[\boldsymbol{H}]\boldsymbol{Q}^T \qquad \text{for all } \boldsymbol{H} \in \text{Lin}, \boldsymbol{Q} \in \mathscr{G}; \tag{2.2.4}$$

and if the body is hyperelastic,

$$\boldsymbol{H} \cdot \mathbf{C}[\boldsymbol{G}] = \boldsymbol{G} \cdot \mathbf{C}[\boldsymbol{H}] \quad \text{for all } \boldsymbol{G}, \boldsymbol{H} \in \text{Lin}. \tag{2.2.5}$$

The first property is a consequence of observer independence. Following GURTIN [1,§29] (cf. DEL PIERO [6,§6]), we show (2.2.2) by taking $\boldsymbol{Q} = \boldsymbol{Q}(\tau)$ in (2.1.9) and differentiating with respect to τ to get

$$D\hat{\boldsymbol{S}}(\boldsymbol{Q}\boldsymbol{F})[\dot{\boldsymbol{Q}}\boldsymbol{F}] = \dot{\boldsymbol{Q}}\hat{\boldsymbol{S}}(\boldsymbol{F}). \tag{2.2.6}$$

The choice

$$\boldsymbol{Q}(\tau) = \mathrm{e}^{\boldsymbol{W}\tau},$$

where $\boldsymbol{W} \in$ Skw, and evaluation of (2.2.6) at $\tau = 0$ lead to

$$D\hat{\boldsymbol{S}}(\boldsymbol{F})[\boldsymbol{W}\boldsymbol{F}] = \boldsymbol{W}\hat{\boldsymbol{S}}(\boldsymbol{F}). \tag{2.2.7}$$

Evaluation of (2.2.7) at $\boldsymbol{F} = \mathbf{1}$ establishes (2.2.2).

The second property follows from the symmetry of the Cauchy stress. By differentiating the third equation of (2.1.7), one finds that

$$D\hat{\boldsymbol{S}}(\boldsymbol{F})[\boldsymbol{A}]\boldsymbol{F}^T + \hat{\boldsymbol{S}}(\boldsymbol{F})\boldsymbol{A}^T = (\det \boldsymbol{F})\operatorname{tr}(\boldsymbol{A}\boldsymbol{F}^{-1})\hat{\boldsymbol{T}}(\boldsymbol{F}) + (\det \boldsymbol{F})D\hat{\boldsymbol{T}}(\boldsymbol{F})[\boldsymbol{A}]. \tag{2.2.8}$$

At $\boldsymbol{F} = \mathbf{1}$ (2.2.8) becomes

$$\mathbf{C}[\boldsymbol{A}] = D\hat{\boldsymbol{T}}(\mathbf{1})[\boldsymbol{A}] - \mathring{\boldsymbol{T}}\boldsymbol{A}^T + (\operatorname{tr}\boldsymbol{A})\mathring{\boldsymbol{T}}. \tag{2.2.9}$$

Both $\mathring{\boldsymbol{T}}$ and $D\hat{\boldsymbol{T}}(\mathbf{1}[\boldsymbol{A}]$ are symmetric, so for symmetric $\boldsymbol{A}$, (2.2.9) implies (2.2.3).

The last two properties are more transparent, and we refer to GURTIN [1,§29] for their proofs.

With linearization in mind, it is natural to introduce the displacement

$$\boldsymbol{u}(\boldsymbol{p}) = \boldsymbol{f}(\boldsymbol{p}) - \boldsymbol{p}. \tag{2.2.10}$$

The displacement gradient

$$\boldsymbol{H} = \nabla \boldsymbol{u} \tag{2.2.11}$$

is related to the deformation gradient by

$$\boldsymbol{F} = \mathbf{1} + \boldsymbol{H}. \tag{2.2.12}$$

We are interested in the case where $\boldsymbol{H}$ is small.

In view of the assumed differentiability of $\hat{\boldsymbol{S}}$,

$$\hat{\boldsymbol{S}}(\boldsymbol{F}) = \hat{\boldsymbol{S}}(\mathbf{1} + \boldsymbol{H}) = \hat{\boldsymbol{S}}(\mathbf{1}) + D\hat{\boldsymbol{S}}(\mathbf{1})[\boldsymbol{H}] + \mathrm{o}(\boldsymbol{H}) \tag{2.2.13}$$

as $\boldsymbol{H} \to \boldsymbol{0}$. Incorporation of (2.1.2), (2.2.2), and (2.2.3) into (2.2.13) yields the asymptotic form of the constitutive relation: *

$$\boldsymbol{S} = \mathring{\boldsymbol{T}} + \boldsymbol{W}\mathring{\boldsymbol{T}} + \tfrac{1}{2}(\boldsymbol{E}\mathring{\boldsymbol{T}} - \mathring{\boldsymbol{T}}\boldsymbol{E}) + \mathbf{L}[\boldsymbol{E}] + \mathrm{o}(\boldsymbol{H}) \tag{2.2.14}$$

as $\boldsymbol{H} \to \boldsymbol{0}$, where

$$\boldsymbol{E} = \tfrac{1}{2}(\boldsymbol{H} + \boldsymbol{H}^T) \tag{2.2.15}$$

and

$$\boldsymbol{W} = \tfrac{1}{2}(\boldsymbol{H} - \boldsymbol{H}^T) \tag{2.2.16}$$

are the infinitesimal strain and infinitesimal rotation, respectively, and $\mathbf{L}$ is the linear transformation from Sym to Sym defined by

$$\mathbf{L}[\boldsymbol{E}] = \operatorname{sym}\mathbf{C}[\boldsymbol{E}]. \tag{2.2.17}$$

We shall refer to $\mathbf{L}$ as the **incremental elasticity tensor** in order to distinguish it from the elasticity tensor $\mathbf{C}$. When the residual stress vanishes and the term $\mathrm{o}(\boldsymbol{H})$ is neglected, (2.2.14) collapses to the stress-strain law of classical linear elasticity.

When the residual stress does not vanish, the stress $\boldsymbol{S}$ is generally not even approximately symmetric. However, calculation of $\boldsymbol{S}\boldsymbol{F}^{\mathrm{T}}$ using (2.2.14) and (2.2.12) and use of the symmetry of $\mathring{\boldsymbol{T}}$ and $\boldsymbol{L}[\boldsymbol{E}]$ establish that

$$\boldsymbol{S}\boldsymbol{F}^T = \boldsymbol{F}\boldsymbol{S}^T + \mathrm{o}(\boldsymbol{H}).$$

Thus *balance of moments is satisfied automatically to within* $\mathrm{o}(\boldsymbol{H})$ *when the constitutive equation* (2.2.14) *is employed* (cf. (2.1.12)).

For $\boldsymbol{Q} \in \mathscr{G}$ it follows from (2.2.4) and (2.2.17) that

$$\mathbf{L}[\boldsymbol{Q}\boldsymbol{E}\boldsymbol{Q}^T] = \boldsymbol{Q}\mathbf{L}[\boldsymbol{E}]\boldsymbol{Q}^T \tag{2.2.18}$$

for all $\boldsymbol{E} \in \mathrm{Sym}$.

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* The same result is obtained by starting with a reduced form of the general elastic constitutive equation which incorporates observer independence.

If the body is hyperelastic, (2.2.5) implies

$$\tilde{\boldsymbol{E}} \cdot \mathbf{C}[\boldsymbol{E}] = \boldsymbol{E} \cdot \mathbf{C}[\tilde{\boldsymbol{E}}]$$

for all $\boldsymbol{E}$, $\tilde{\boldsymbol{E}} \in$ Sym. Combining this with

$$\mathbf{C}[\boldsymbol{E}] = \tfrac{1}{2}(\boldsymbol{E}\mathring{\boldsymbol{T}} - \mathring{\boldsymbol{T}}\boldsymbol{E}) + \mathbf{L}[\boldsymbol{E}]$$

(cf. (2.2.3) and (2.2.17)) and noting the symmetry of $\mathring{\boldsymbol{T}}$, we are lead to

$$\tilde{\boldsymbol{E}} \cdot \mathbf{L}[\boldsymbol{E}] = \boldsymbol{E} \cdot \mathbf{L}[\tilde{\boldsymbol{E}}] \tag{2.2.19}$$

for all $\boldsymbol{E}$, $\tilde{\boldsymbol{E}} \in$ Sym.

The constitutive equation (2.2.14) or its equivalent has been derived and used by various investigators. BAZANT [7] has shown that the formulations of SOUTHWELL, BIEZENO and HENCKY, TREFFTZ, and BIOT all result in a constitutive equation that is essentially (2.2.14). Also, it is well known that the constitutive equation for small elastic deformations superimposed on large takes the same form as (2.2.14); for example see [8], [9] and [10]; but our $\mathring{\boldsymbol{T}}$ need not be the result of an elastic deformation.

2.3. Basic equations of the linear theory

Here we collect together the results of the previous subsections that are pertinent to the linearized theory. Discarding terms of order $o(\boldsymbol{H})$ in the constitutive equation (2.2.14), we obtain the **basic field equations of linear elastostatics with residual stress:**

$$\left.\begin{aligned} &\boldsymbol{H} = \nabla \boldsymbol{u}, \\ &\boldsymbol{E} = \tfrac{1}{2}(\boldsymbol{H} + \boldsymbol{H}^T), \\ &\boldsymbol{W} = \tfrac{1}{2}(\boldsymbol{H} - \boldsymbol{H}^T), \\ &\boldsymbol{S} = \mathring{\boldsymbol{T}} + \boldsymbol{W}\mathring{\boldsymbol{T}} + \tfrac{1}{2}(\boldsymbol{E}\mathring{\boldsymbol{T}} - \mathring{\boldsymbol{T}}\boldsymbol{E}) + \mathbf{L}[\boldsymbol{E}], \\ &\nabla \cdot \boldsymbol{S} + \boldsymbol{b} = \boldsymbol{0}. \end{aligned}\right\} \tag{2.3.1}$$

These equations hold in B. The residual stress meets

$$\nabla \cdot \mathring{\boldsymbol{T}} = \boldsymbol{0}, \quad \mathring{\boldsymbol{T}}^T = \mathring{\boldsymbol{T}} \tag{2.3.2}$$

in B, and

$$\mathring{\boldsymbol{T}}\boldsymbol{n} = \boldsymbol{0} \tag{2.3.3}$$

on ∂B, where $\boldsymbol{n}$ is the outward unit normal to ∂B.

Experiments for the determination of the incremental elasticities and the residual stress in right circular cylinders are suggested in Subsections 3.4 and 3.5. As usual, the experiments correspond to certain boundary value problems for the system (2.3.1) (with $\mathbf{L}$ and $\mathring{\boldsymbol{T}}$ viewed as given) whose solutions are known analytically. Of course, this procedure is specious if these boundary value problems do not possess unique solutions which depend continuously on the data. This issue has been resolved by CARLSON and MAN [11]; it is enough that the incremental elasticity tensor field $\mathbf{L}$ be positive definite and that the residual stress field $\mathring{\boldsymbol{T}}$ be sufficiently small.

If $\boldsymbol{Q} \in \mathrm{Orth}^+$ is a member of the material symmetry group of the body, then we must have both

$$\boldsymbol{Q}\mathring{\boldsymbol{T}} = \mathring{\boldsymbol{T}}\boldsymbol{Q} \tag{2.3.4}$$

and

$$\mathsf{L}[\boldsymbol{Q}\boldsymbol{E}\boldsymbol{Q}^T] = \boldsymbol{Q}\mathsf{L}[\boldsymbol{E}]\boldsymbol{Q}^T \tag{2.3.5}$$

for all $\boldsymbol{E} \in$ Sym.

If the body is hyperelastic, then the incremental elasticity tensor L *has the major symmetry*

$$\tilde{\boldsymbol{E}} \cdot \mathsf{L}[\boldsymbol{E}] = \boldsymbol{E} \cdot \mathsf{L}[\tilde{\boldsymbol{E}}] \tag{2.3.6}$$

for all $\boldsymbol{E}$, $\tilde{\boldsymbol{E}} \in$ Sym.

We end this section with an observation on the traction boundary condition. In view of (2.3.3), the traction on the boundary computed from (2.3.1d) is

$$\boldsymbol{o} = \boldsymbol{S}\boldsymbol{n} = \left(-\tfrac{1}{2}\mathring{\boldsymbol{T}}\boldsymbol{E} + \mathsf{L}[\boldsymbol{E}]\right)\boldsymbol{n}. \tag{2.3.7}$$

Physically, $\boldsymbol{o} = \boldsymbol{S}\boldsymbol{n}$ represents the force per unit of undeformed area acting at the corresponding point of the deformed surface. The force per unit of deformed area there is

$$\boldsymbol{\ell} = \boldsymbol{T}\bar{\boldsymbol{n}}, \tag{2.3.8}$$

where $\bar{\boldsymbol{n}}$ is the outward unit normal to the deformed surface. By (2.1.7) and (2.2.12), the Cauchy stress can be written as

$$\boldsymbol{T} = \boldsymbol{S} + \mathring{\boldsymbol{T}}\boldsymbol{H}^T - (\operatorname{tr}\boldsymbol{E})\mathring{\boldsymbol{T}} + \mathrm{o}(\boldsymbol{H}),$$

so by using (2.2.14) we find that

$$\boldsymbol{T} = \mathring{\boldsymbol{T}} + \boldsymbol{W}\mathring{\boldsymbol{T}} - \mathring{\boldsymbol{T}}\boldsymbol{W} + \tfrac{1}{2}(\boldsymbol{E}\mathring{\boldsymbol{T}} + \mathring{\boldsymbol{T}}\boldsymbol{E}) - (\operatorname{tr}\boldsymbol{E})\mathring{\boldsymbol{T}} + \mathsf{L}[\boldsymbol{E}] + \mathrm{o}(\boldsymbol{H}).$$

Since $\bar{\boldsymbol{n}}$ is related to $\boldsymbol{n}$ by (see [1,§6])

$$\bar{\boldsymbol{n}} = \left(\boldsymbol{n} \cdot \boldsymbol{F}^{-1}\boldsymbol{F}^{-T}\boldsymbol{n}\right)^{-1/2}\boldsymbol{F}^{-T}\boldsymbol{n},$$

we have

$$\bar{\boldsymbol{n}} = \boldsymbol{n} - \boldsymbol{H}^T\boldsymbol{n} + (\boldsymbol{n} \cdot \boldsymbol{E}\boldsymbol{n})\boldsymbol{n} + \mathrm{o}(\boldsymbol{H}),$$

and $\boldsymbol{T}\bar{\boldsymbol{n}}$ can be calculated to show that the traction vectors $\boldsymbol{o}$ and $\boldsymbol{\ell}$ satisfy

$$\boldsymbol{o} = \boldsymbol{\ell} + \mathrm{o}(\boldsymbol{H}). \tag{2.3.9}$$

Thus, *to within the accuracy of the linearized theory,* $\boldsymbol{S}\boldsymbol{n}$ *may be interpreted as supplying the Cauchy traction.*

3. Application to right circular cylinders

As previously noted, the residual stress in a body is closely associated to the body geometry by the requirement that $\mathring{\boldsymbol{T}}\boldsymbol{n} = \boldsymbol{0}$ on ∂B. For this reason the body shape must be specified in order to obtain further results. Here we take the undeformed body to be a right circular cylinder, choose a particular class of residual stress *, and assume that the incremental elasticity tensor has a homogeneous and isotropic structure. We then consider two universal solutions which can be realized in the laboratory and thereby

* For the most part we work with $\mathring{\boldsymbol{T}} = \mathring{\boldsymbol{T}}(r, \theta)$ and its special case $\mathring{\boldsymbol{T}} = \mathring{\boldsymbol{T}}(r)$. A few observations are made on the $\mathring{\boldsymbol{T}} = \mathring{\boldsymbol{T}}(\theta, z)$ and $\mathring{\boldsymbol{T}} = \mathring{\boldsymbol{T}}(r, z)$ cases. Here r, θ, and z are cylindrical coordinates appropriate to the given undeformed shape.

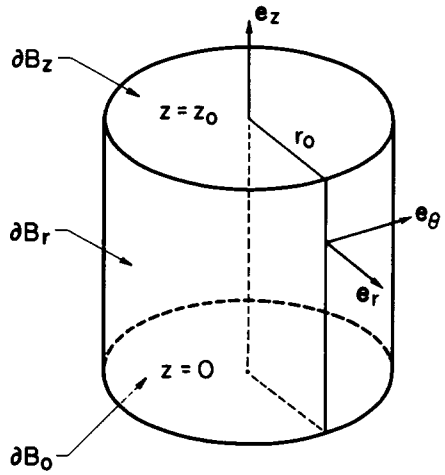


Figure 1. The cylinder shown in the reference configuration.

provide a means for the experimental determination of the incremental elasticities and the residual stress.

3.1. The cylinder

Let the unloaded but residually stressed body B be a right circular cylinder of radius r_0 with generators parallel to the z axis. A cylindrical coordinate system will be used, with respect to which the ends of the cylinder, ∂B_z and ∂B_0, are located at $z = z_0$ and $z = 0$, respectively. The lateral surface is denoted by ∂B_r. The body is pictured in Figure 1.

The equilibrium equations and surface tractions for both the unloaded and loaded problems are displayed in cylindrical components in this section with all vectors and tensors written in terms of their physical components. The body force will be taken to be zero.

For the unloaded cylinder the components of the equilibrium equation (2.3.2a) for the residual stress are

$$\left.\begin{aligned} &\frac{\partial \mathring{T}_{rr}}{\partial r} + \frac{1}{r}\frac{\partial \mathring{T}_{r\theta}}{\partial \theta} + \frac{\partial \mathring{T}_{rz}}{\partial z} + \frac{1}{r}(\mathring{T}_{rr} - \mathring{T}_{\theta\theta}) = 0, \\ &\frac{\partial \mathring{T}_{r\theta}}{\partial r} + \frac{1}{r}\frac{\partial \mathring{T}_{\theta\theta}}{\partial \theta} + \frac{\partial \mathring{T}_{\theta z}}{\partial z} + \frac{2}{r}\mathring{T}_{r\theta} = 0, \\ &\frac{\partial \mathring{T}_{rz}}{\partial r} + \frac{1}{r}\frac{\partial \mathring{T}_{\theta z}}{\partial \theta} + \frac{\partial \mathring{T}_{zz}}{\partial z} + \frac{1}{r}\mathring{T}_{rz} = 0, \end{aligned}\right\} \tag{3.1.1}$$

where the symmetry condition (2.3.2b) has been incorporated. The boundary condition (2.3.3) is

$$\begin{bmatrix} \mathring{T}_{rz}(r, \theta, z_0) \\ \mathring{T}_{\theta z}(r, \theta, z_0) \\ \mathring{T}_{zz}(r, \theta, z_0) \end{bmatrix} = [\boldsymbol{0}] \quad \text{and} \quad \begin{bmatrix} \mathring{T}_{rz}(r, \theta, 0) \\ \mathring{T}_{\theta z}(r, \theta, 0) \\ \mathring{T}_{zz}(r, \theta, 0) \end{bmatrix} = [\boldsymbol{0}] \tag{3.1.2}$$

on ∂B_z and ∂B_0, respectively, and

$$\begin{bmatrix} \mathring{T}_{rr}(r_0, \theta, z) \\ \mathring{T}_{r\theta}(r_0, \theta, z) \\ \mathring{T}_{rz}(r_0, \theta, z) \end{bmatrix} = [\mathbf{0}] \tag{3.1.3}$$

on ∂B_r.

The components of the equilibrium equation (2.3.1e) for the deformed body are

$$\left.\begin{aligned} &\frac{\partial S_{rr}}{\partial r} + \frac{1}{r}\frac{\partial S_{r\theta}}{\partial \theta} + \frac{\partial S_{rz}}{\partial z} + \frac{1}{r}(S_{rr} - S_{\theta\theta}) = 0, \\ &\frac{\partial S_{\theta r}}{\partial r} + \frac{1}{r}\frac{\partial S_{\theta\theta}}{\partial \theta} + \frac{\partial S_{\theta z}}{\partial z} + \frac{1}{r}(S_{r\theta} + S_{\theta r}) = 0, \\ &\frac{\partial S_{zr}}{\partial r} + \frac{1}{r}\frac{\partial S_{z\theta}}{\partial \theta} + \frac{\partial S_{zz}}{\partial z} + \frac{1}{r}S_{zr} = 0. \end{aligned}\right\} \tag{3.1.4}$$

The tractions on the boundary of the deformed body can be computed from (2.3.7):

$$[\boldsymbol{S}\boldsymbol{e}_z] = -\tfrac{1}{2}\begin{bmatrix} \mathring{T}_{rr}E_{rz} + \mathring{T}_{r\theta}E_{\theta z} \\ \mathring{T}_{r\theta}E_{rz} + \mathring{T}_{\theta\theta}E_{\theta z} \\ 0 \end{bmatrix} + [\mathbf{L}[\boldsymbol{E}]\boldsymbol{e}_z] \tag{3.1.5}$$

on ∂B_z;

$$[\boldsymbol{S}(-\boldsymbol{e}_z)] = \tfrac{1}{2}\begin{bmatrix} \mathring{T}_{rr}E_{rz} + \mathring{T}_{r\theta}E_{\theta z} \\ \mathring{T}_{r\theta}E_{rz} + \mathring{T}_{\theta\theta}E_{\theta z} \\ 0 \end{bmatrix} - [\mathbf{L}[\boldsymbol{E}]\boldsymbol{e}_z] \tag{3.1.6}$$

on ∂B_0; and

$$[\boldsymbol{S}\boldsymbol{e}_r] = -\tfrac{1}{2}\begin{bmatrix} 0 \\ \mathring{T}_{\theta\theta}E_{r\theta} + \mathring{T}_{\theta z}E_{rz} \\ \mathring{T}_{\theta z}E_{r\theta} + \mathring{T}_{zz}E_{rz} \end{bmatrix} + [\mathbf{L}[\boldsymbol{E}]\boldsymbol{e}_r] \tag{3.1.7}$$

on ∂B_r, where (3.1.2) and (3.1.3) have been used.

For convenience, we make simplifying assumptions concerning the forms of the incremental elasticity tensor and the residual stress. It will be assumed that $\mathbf{L}$ is isotropic and homogeneous, so it can be represented as

$$\mathbf{L}[\boldsymbol{E}] = 2\bar{\mu}\boldsymbol{E} + \bar{\lambda}(\operatorname{tr}\boldsymbol{E})\mathbf{1}. \tag{3.1.8}$$

The constants $\bar{\mu}$ and $\bar{\lambda}$ are the counterparts of the Lamé constants of the classical theory. Of course, whether or not this assumption is appropriate would depend on the particular material. It is important to realize that it is not assumed that the overall response of the body is isotropic. Indeed, it was shown in [5] that if the response of an elastic body of any shape is isotropic, then necessarily $\mathring{\boldsymbol{T}} \equiv \boldsymbol{0}$. In all that follows, the assumption that $\mathbf{L}[\boldsymbol{E}]$ has the form (3.1.8) is taken to be inviolable, and conclusions concerning the possible forms of the residual stress should be interpreted with this in mind. There is no conceptual difficulty with taking $\mathbf{L}[\boldsymbol{E}]$ to be more general, but it would complicate the subsequent analysis.

Sometimes enough is known about the manufacture of a body to permit reasonable assumptions about the general form of its residual stress. For example, one would

expect the residual stress in drawn wire to depend on only the radial coordinate. For the remainder of this section we will assume primarily that $\mathring{\boldsymbol{T}} = \mathring{\boldsymbol{T}}(\mathrm{r}, \theta)$ and make only occasional reference to other special forms.

3.2. A priori results on the residual stress

In the context of a specific body shape, the equilibrium equations and boundary conditions are often strong enough to delimit substantially a residual stress field already assumed to have particular structure. We explore this idea for right circular cylinders. The results obtained in this subsection are *universal* in the sense that they are independent of any constitutive assumptions for the response of the body to deformations from the reference configuration.

The special forms considered for the residual stress are where $\mathring{\boldsymbol{T}}$ depends only on θ and z, r and θ, r, and r and z.

Residual stress depending on θ and z alone

When the residual stress depends on only the variables θ and z, the boundary conditions (3.1.2a) and (3.1.3) become

$$\begin{bmatrix} \mathring{T}_{rz}(\theta, z_0) \\ \mathring{T}_{\theta z}(\theta, z_0) \\ \mathring{T}_{zz}(\theta, z_0) \end{bmatrix} = [\boldsymbol{0}] \tag{3.2.1}$$

on ∂B_z, and

$$\begin{bmatrix} \mathring{T}_{rr}(\theta, z) \\ \mathring{T}_{r\theta}(\theta, z) \\ \mathring{T}_{rz}(\theta, z) \end{bmatrix} = [\boldsymbol{0}] \tag{3.2.2}$$

on ∂B_r. Because there is no radial dependence, (3.2.2) holds throughout the cylinder, so each of the components $\mathring{T}_{rr}$, $\mathring{T}_{r\theta}$, and $\mathring{T}_{rz}$ is identically zero. The equilibrium equations (3.1.1) become

$$\left.\begin{aligned} &\mathring{T}_{\theta\theta} = 0, \\ &\frac{\partial \mathring{T}_{\theta z}}{\partial z} = 0, \\ &\frac{\partial \mathring{T}_{zz}}{\partial z} = 0. \end{aligned}\right\} \tag{3.2.3}$$

The last two equations of (3.2.3) imply that $\mathring{T}_{\theta z}$ and $\mathring{T}_{zz}$ are functions of θ only, so by the boundary condition (3.2.1), they must vanish throughout the body. Therefore *no nonzero residual stress depending on only θ and z, or either θ or z alone, can exist in a right circular cylinder.*

Residual stress depends on r and θ alone

When the residual stress depends on only the variables r and θ, the boundary conditions ∂B_z and ∂B_0 become

$$\begin{bmatrix} \mathring{T}_{rz}(r, \theta) \\ \mathring{T}_{\theta z}(r, \theta) \\ \mathring{T}_{zz}(r, \theta) \end{bmatrix} = [\boldsymbol{0}]. \tag{3.2.4}$$

Because there is no z dependence, these components of the residual stress must be zero throughout the body. Hence *a state of residual stress depending on only r and θ is plane,* and $\mathring{\boldsymbol{T}}$ can be represented by an Airy stress function ϕ as

$$[\mathring{\boldsymbol{T}}] = \begin{bmatrix} \left(\frac{1}{r}\phi_{,r} + \frac{1}{r^2}\phi_{,\theta\theta}\right) & \left(\frac{1}{r^2}\phi_{,\theta} - \frac{1}{r}\phi_{,\theta r}\right) & 0 \\ \left(\frac{1}{r^2}\phi_{,\theta} - \frac{1}{r}\phi_{,\theta r}\right) & \phi_{,rr} & 0 \\ 0 & 0 & 0 \end{bmatrix}. \tag{3.2.5}$$

In terms of the stress function, the remaining boundary condition is

$$\begin{bmatrix} \frac{1}{r_0}\phi_{,r}(r_0, \theta) + \frac{1}{r_0^2}\phi_{,\theta\theta}(r_0, \theta) \\ \frac{1}{r_0^2}\phi_{,\theta}(r_0, \theta) - \frac{1}{r_0}\phi_{,r\theta}(r_0, \theta) \\ 0 \end{bmatrix} = [\boldsymbol{0}]. \tag{3.2.6}$$

Residual stress depends on r alone

In this case it follows from the previous paragraph that $\mathring{\boldsymbol{T}}$ is plane, but in addition the equilibrium equation (3.1.1b) reduces to

$$\frac{d}{dr}\mathring{T}_{r\theta} + \frac{2}{r}\mathring{T}_{r\theta} = 0.$$

Therefore, since $\mathring{T}_{r\theta}(r_0) = 0$, $\mathring{T}_{r\theta} \equiv 0$. Hence *a state of residual stress depending on only r has at most nonzero components* $\mathring{T}_{rr}$ *and* $\mathring{T}_{\theta\theta}$, and $\mathring{\boldsymbol{T}}$ can be represented through an Airy stress function $\phi = \phi(r)$ as

$$[\mathring{\boldsymbol{T}}] = \begin{bmatrix} \frac{1}{r}\phi_{,r} & 0 & 0 \\ 0 & \phi_{,rr} & 0 \\ 0 & 0 & 0 \end{bmatrix}. \tag{3.2.7}$$

The boundary condition on ∂B_r requires that

$$\phi_{,r}(r_0) = 0. \tag{3.2.8}$$

Residual stress depending on r and z alone

When the residual stress depends on r and z alone, the equilibrium equations for the unloaded problem become

$$\left.\begin{aligned} &\frac{\partial \mathring{T}_{rr}}{\partial r} + \frac{\partial \mathring{T}_{rz}}{\partial z} + \frac{1}{r}(\mathring{T}_{rr} - \mathring{T}_{\theta\theta}) = 0, \\ &\frac{\partial \mathring{T}_{r\theta}}{\partial r} + \frac{\partial \mathring{T}_{\theta z}}{\partial z} + \frac{2}{r}\mathring{T}_{r\theta} = 0, \\ &\frac{\partial \mathring{T}_{rz}}{\partial r} + \frac{\partial \mathring{T}_{zz}}{\partial z} + \frac{1}{r}\mathring{T}_{rz} = 0. \end{aligned}\right\} \tag{3.2.9}$$

The boundary conditions are

$$\begin{bmatrix} \mathring{T}_{rz}(r, z_0) \\ \mathring{T}_{\theta z}(r, z_0) \\ \mathring{T}_{zz}(r, z_0) \end{bmatrix} = [\mathbf{0}] \quad \text{and} \quad \begin{bmatrix} \mathring{T}_{rz}(r, 0) \\ \mathring{T}_{\theta z}(r, 0) \\ \mathring{T}_{zz}(r, 0) \end{bmatrix} = [\mathbf{0}] \tag{3.2.10}$$

on ∂B_z and ∂B_0, respectively, and

$$\begin{bmatrix} \mathring{T}_{rr}(r_0, z) \\ \mathring{T}_{r\theta}(r_0, z) \\ \mathring{T}_{rz}(r_0, z) \end{bmatrix} = [\mathbf{0}] \tag{3.2.11}$$

on ∂B_r. The unloaded problem by itself leads to no useful restrictions on the form of the residual stress.

A complete stress function solution of the equilibrium equations (3.2.9) in the cylinder is given by

$$[\mathring{\boldsymbol{T}}] = \begin{bmatrix} \frac{1}{r}(\psi - \chi_{,zz}) & \frac{1}{r^2}\omega_{,z} & \frac{1}{r}\chi_{,rz} \\ \frac{1}{r^2}\omega_{,z} & \psi_{,r} & -\frac{1}{r^2}\omega_{,r} \\ \frac{1}{r}\chi_{,rz} & -\frac{1}{r^2}\omega_{,r} & -\frac{1}{r}\chi_{,rr} \end{bmatrix}, \tag{3.2.12}$$

where ψ, χ, and ω are sufficiently smooth functions of r and z.

Even in the important special case of torsionless axisymmetry where $\mathring{T}_{r\theta} \equiv \mathring{T}_{\theta z} \equiv 0$, we find no useful restrictions on the residual stress. In this case the stress function solution (3.2.12) with $\omega \equiv 0$ retains its completeness.

3.3 Universal solutions

Especially important in the design of experiments for the determination of material properties are **universal solutions,** *which are displacement fields that can be maintained in any body of a particular constitutive class by surface tractions alone.* Here the constitutive class is defined by the assumptions that B is the right circular cylinder * described in Subsection 3.1, the incremental elasticity tensor $\mathbf{L}$ has the homogeneous and isotropic structure $\mathbf{L}[\boldsymbol{E}] = 2\bar{\mu}\boldsymbol{E} + \bar{\lambda}(\text{tr } \boldsymbol{E})\mathbf{1}$, and the residual stress $\mathring{\boldsymbol{T}}$ has one of the special forms considered in Subsection 3.2.

Residual stress depends on r and θ alone

In this case the residual stress $\mathring{\boldsymbol{T}}$ is given in terms of the stress function ϕ by (3.2.5). Then with (2.3.1d) and (3.1.8), the three components of the equation of equilibrium (3.1.4) are

$$2\left[\frac{\partial E_{rr}}{\partial r} + \frac{1}{r}\frac{\partial E_{r\theta}}{\partial \theta} + \frac{\partial E_{rz}}{\partial z} + \frac{1}{r}(E_{rr} - E_{\theta\theta})\right]\bar{\mu} + \left[\frac{\partial E_{rr}}{\partial r} + \frac{\partial E_{\theta\theta}}{\partial r} + \frac{\partial E_{zz}}{\partial r}\right]\bar{\lambda}$$
$$+ \frac{1}{r}\left[-\frac{1}{2r}\frac{\partial E_{r\theta}}{\partial \theta} - \frac{1}{2}\frac{\partial E_{rz}}{\partial z}\right]\phi_{,r} + \frac{1}{r}\left[\frac{\partial W_{r\theta}}{\partial \theta} + \frac{1}{2}\frac{\partial E_{r\theta}}{\partial \theta}\right]\phi_{,rr}$$

* Since we are viewing the residual stress field as constitutive information, the shape of the undeformed body must be specified.

$$+\frac{1}{r^2}\left[\frac{\partial W_{r\theta}}{\partial r}-\frac{1}{r}\left(\frac{\partial E_{\theta\theta}}{\partial\theta}-\frac{\partial E_{rr}}{\partial\theta}\right)-\frac{1}{2}\frac{\partial E_{\theta z}}{\partial z}\right]\phi_{,\theta}$$

$$+\frac{1}{r}\left[-\frac{\partial W_{r\theta}}{\partial r}-\frac{1}{2r}E_{r\theta}+\frac{1}{r}\left(\frac{\partial E_{\theta\theta}}{\partial\theta}-\frac{\partial E_{rr}}{\partial\theta}\right)+\frac{1}{2}\frac{\partial E_{\theta z}}{\partial z}\right]\phi_{,r\theta}$$

$$+\frac{1}{r^2}\left[-\frac{1}{2r}\frac{\partial E_{r\theta}}{\partial\theta}-\frac{1}{r}(E_{\theta\theta}-E_{rr})-\frac{1}{2}\frac{\partial E_{rz}}{\partial z}\right]\phi_{,\theta\theta}$$

$$-\frac{1}{2r^3}[E_{r\theta}]\phi_{,\theta\theta\theta}+\frac{1}{2r}[E_{r\theta}]\phi_{,rr\theta}+\frac{1}{r^2}[E_{\theta\theta}-E_{rr}]\phi_{,r\theta\theta}=0,$$

$$2\left[\frac{\partial E_{r\theta}}{\partial r}+\frac{1}{r}\frac{\partial E_{\theta\theta}}{\partial\theta}+\frac{\partial E_{\theta z}}{\partial z}+\frac{2}{r}E_{r\theta}\right]\bar{\mu}+\frac{1}{r}\left[\frac{\partial E_{rr}}{\partial\theta}+\frac{\partial E_{\theta\theta}}{\partial\theta}+\frac{\partial E_{zz}}{\partial\theta}\right]\bar{\lambda}$$

$$+\frac{1}{r}\left[-\frac{\partial W_{r\theta}}{\partial r}+\frac{1}{2}\frac{\partial E_{r\theta}}{\partial r}-\frac{1}{2r}E_{r\theta}\right]\phi_{,r}+\left[-\frac{1}{2}\frac{\partial E_{\theta z}}{\partial z}-\frac{1}{2}\frac{\partial E_{r\theta}}{\partial r}+\frac{1}{2r}E_{r\theta}\right]\phi_{,rr}$$

$$+\frac{1}{r^2}\left[-\frac{1}{2}\frac{\partial E_{rz}}{\partial z}-\frac{1}{r}\frac{\partial W_{r\theta}}{\partial\theta}+\frac{2}{r}(E_{rr}-E_{\theta\theta})-\left(\frac{\partial E_{rr}}{\partial r}-\frac{\partial E_{\theta\theta}}{\partial r}\right)\right]\phi_{,\theta}$$

$$+\frac{1}{r}\left[\frac{1}{r}\frac{\partial W_{r\theta}}{\partial\theta}-\frac{2}{r}(E_{rr}-E_{\theta\theta})+\left(\frac{\partial E_{rr}}{\partial r}-\frac{\partial E_{\theta\theta}}{\partial r}\right)+\frac{1}{2}\frac{\partial E_{rz}}{\partial z}\right]\phi_{,r\theta}$$

$$+\frac{1}{r}\left[-\frac{\partial W_{r\theta}}{\partial r}+\frac{1}{2}\frac{\partial E_{r\theta}}{\partial r}-\frac{1}{r}E_{r\theta}\right]\phi_{,\theta\theta}-\tfrac{1}{2}[E_{r\theta}]\phi_{,rrr}+\frac{1}{2r^2}[E_{r\theta}]\phi_{,\theta\theta r}$$

$$+\frac{1}{r}[E_{rr}-E_{\theta\theta}]\phi_{,rr\theta}=0,$$

$$2\left[\frac{\partial E_{rz}}{\partial r}+\frac{1}{r}\frac{\partial E_{\theta z}}{\partial\theta}+\frac{\partial E_{zz}}{\partial z}+\frac{1}{r}E_{rz}\right]\bar{\mu}+\left[\frac{\partial E_{rr}}{\partial z}+\frac{\partial E_{\theta\theta}}{\partial z}+\frac{\partial E_{zz}}{\partial z}\right]\bar{\lambda}$$

$$+\frac{1}{r}\left[-\frac{\partial W_{rz}}{\partial r}+\frac{1}{2}\frac{\partial E_{rz}}{\partial r}\right]\phi_{,r}+\frac{1}{r^2}\left[-\frac{\partial W_{rz}}{\partial r}+\frac{1}{2}\frac{\partial E_{rz}}{\partial r}\right]\phi_{,\theta\theta}$$

$$+\frac{1}{r}\left[-W_{rz}+\tfrac{1}{2}E_{rz}-\frac{\partial W_{\theta z}}{\partial\theta}+\frac{1}{2}\frac{\partial E_{\theta z}}{\partial\theta}\right]\phi_{,rr}$$

$$+\frac{1}{r^2}\left[-\frac{\partial W_{\theta z}}{\partial r}+\frac{1}{2}\frac{\partial E_{\theta z}}{\partial r}-\tfrac{1}{2}E_{\theta z}-\frac{1}{r}\frac{\partial W_{rz}}{\partial\theta}+\frac{1}{2r}\frac{\partial E_{rz}}{\partial\theta}+\frac{1}{r}W_{\theta z}\right]\phi_{,\theta}$$

$$+\frac{1}{r}\left[\frac{\partial W_{\theta z}}{\partial r}-\frac{1}{r}W_{\theta z}-\frac{1}{2}\frac{\partial E_{\theta z}}{\partial r}+\frac{1}{2r}E_{\theta z}+\frac{1}{r}\frac{\partial W_{rz}}{\partial\theta}-\frac{1}{2r}\frac{\partial E_{rz}}{\partial\theta}\right]\phi_{,\theta r}=0.$$

In order that these equations be satisfied independently of the incremental elastic constants and the residual stress, the expressions in brackets must all vanish. This condition is met if and only if the strain and rotation satisfy the following equations:

$$E_{r\theta}=0, \tag{3.3.1}$$

$$E_{rr}=E_{\theta\theta}, \tag{3.3.2}$$

$$\frac{\partial E_{rr}}{\partial r}=\frac{\partial E_{rr}}{\partial\theta}=0, \tag{3.3.3}$$

$$\frac{\partial E_{zz}}{\partial r}=\frac{\partial E_{zz}}{\partial\theta}=0, \tag{3.3.4}$$

$$\frac{\partial E_{rz}}{\partial z} = \frac{\partial E_{\theta z}}{\partial z} = 0, \tag{3.3.5}$$

$$\frac{\partial W_{r\theta}}{\partial r} = \frac{\partial W_{r\theta}}{\partial \theta} = 0, \tag{3.3.6}$$

$$2\frac{\partial E_{rr}}{\partial z} + \frac{\partial E_{zz}}{\partial z} = 0, \tag{3.3.7}$$

$$\frac{1}{2}\frac{\partial E_{rz}}{\partial r} - \frac{\partial W_{rz}}{\partial r} = 0, \tag{3.3.8}$$

$$\frac{\partial E_{rz}}{\partial r} + \frac{1}{r}\frac{\partial E_{\theta z}}{\partial \theta} + \frac{\partial E_{zz}}{\partial z} + \frac{1}{r}E_{rz} = 0, \tag{3.3.9}$$

$$\left(\tfrac{1}{2}E_{rz} - W_{rz}\right) + \left(\frac{1}{2}\frac{\partial E_{\theta z}}{\partial \theta} - \frac{\partial W_{\theta z}}{\partial \theta}\right) = 0, \tag{3.3.10}$$

$$\frac{1}{r}\left(\tfrac{1}{2}E_{\theta z} - W_{\theta z}\right) + \frac{1}{r}\left(\frac{\partial W_{rz}}{\partial \theta} - \frac{1}{2}\frac{\partial E_{rz}}{\partial \theta}\right) + \left(\frac{\partial W_{\theta z}}{\partial r} - \frac{1}{2}\frac{\partial E_{\theta z}}{\partial r}\right) = 0. \tag{3.3.11}$$

These conditions can be written in terms of derivatives of the displacement by combining them with the strain-displacement and the rotation-displacement equations (2.3.1b) and (2.3.1c). The resulting differential equations in the components of displacement can then be integrated to yield the form for universal solutions since the cylinder is convex. The steps are outlined below.

For convenient reference, the equations relating the displacement to the strain and rotation are displayed here in cylindrical coordinates:

$$\left.\begin{aligned}
&\frac{\partial u_r}{\partial r} = E_{rr},\\
&\frac{1}{r}\left(\frac{\partial u_r}{\partial \theta} - u_\theta\right) = E_{r\theta} + W_{r\theta},\\
&\frac{\partial u_r}{\partial z} = E_{rz} + W_{rz},\\
&\frac{\partial u_\theta}{\partial r} = E_{r\theta} - W_{r\theta},\\
&\frac{1}{r}\left(\frac{\partial u_\theta}{\partial \theta} + u_r\right) = E_{\theta\theta},\\
&\frac{\partial u_\theta}{\partial z} = E_{\theta z} + W_{\theta z},\\
&\frac{\partial u_z}{\partial r} = E_{rz} - W_{rz},\\
&\frac{1}{r}\frac{\partial u_z}{\partial \theta} = E_{\theta z} - W_{\theta z},\\
&\frac{\partial u_z}{\partial z} = E_{zz}.
\end{aligned}\right\} \tag{3.3.12}$$

Conditions (3.3.3), (3.3.4), (3.3.5), and (3.3.6), respectively, imply that

$$E_{rr} = \alpha(z), \tag{3.3.13}$$

$$E_{zz} = \beta(z), \tag{3.3.14}$$

$$E_{rz} = \gamma(r, \theta) \quad \text{and} \quad E_{\theta z} = \lambda(r, \theta), \tag{3.3.15}$$

$$W_{r\theta} = \mu(z). \tag{3.3.16}$$

Equations (3.3.12a) and (3.3.13) imply that

$$u_r(r, \theta, z) = u_r(r_0, \theta, z) + (r - r_0)\alpha(z). \tag{3.3.17}$$

By (3.3.12e), (3.3.2), and (3.3.13),

$$u_\theta(r, \theta, z) = u_\theta(r, \theta_0, z) + r_0(\theta - \theta_0)\alpha(z) - \int_{\theta_0}^{\theta} u_r(r_0, \theta, z)\, d\theta, \tag{3.3.18}$$

where θ_0 is an arbitrary fixed value of θ. Equations (3.3.1) and (3.3.16) can be combined with (3.3.12d) and (3.3.12b) to give

$$u_\theta(r, \theta, z) = u_\theta(r_0, \theta, z) - \mu(z)(r - r_0) \tag{3.3.19}$$

and

$$u_r(r, \theta, z) = u_r(r, \theta_0, z) + r\mu(z)(\theta - \theta_0) + \int_{\theta_0}^{\theta} u_\theta(r, \theta, z)\, d\theta, \tag{3.3.20}$$

respectively.

Equations (3.3.17), (3.3.18), and (3.3.19) together yield

$$u_r(r_0, \theta, z) + (r - r_0)\alpha(z) = u_r(r, \theta_0, z) + r_0(\theta - \theta_0)\mu(z) + \int_{\theta_0}^{\theta} u_\theta(r_0, \theta, z)\, d\theta,$$

which can be differentiated with respect to θ to give

$$u_\theta(r_0, \theta, z) = -r_0\mu(z) + \frac{\partial u_r}{\partial \theta}(r_0, \theta, z). \tag{3.3.21}$$

Using (3.3.18) and (3.3.19), one finds

$$u_\theta(r_0, \theta, z) - (r - r_0)\mu(z) = u_\theta(r, \theta_0, z) + r_0(\theta - \theta_0)\alpha(z) - \int_{\theta_0}^{\theta} u_r(r_0, \theta, z)\, d\theta.$$

Differentiation with respect to θ leads to

$$u_r(r_0, \theta, z) = r_0\alpha(z) - \frac{\partial u_\theta}{\partial \theta}(r_0, \theta, z). \tag{3.3.22}$$

Insertion of (3.3.21) into (3.3.22) provides the differential equation

$$\frac{\partial^2 u_r}{\partial \theta^2}(r_0, \theta, z) + u_r(r_0, \theta, z) = r_0\alpha(z),$$

whose solution,

$$u_r(r_0, \theta, z) = A(z)\cos\theta + B(z)\sin\theta + r_0\alpha(z), \tag{3.3.23}$$

can be used in (3.3.17) to get

$$u_r(r, \theta, z) = A(z)\cos\theta + B(z)\sin\theta + r\alpha(z). \tag{3.3.24}$$

By substitution of (3.3.23) into (3.3.21), we find

$$u_\theta(r_0, \theta, z) = -r_0\mu(z) - A(z)\sin\theta + B(z)\cos\theta,$$

which, together with (3.3.19), gives

$$u_\theta(r, \theta, z) = -r\mu(z) - A(z)\sin\theta + B(z)\cos\theta. \tag{3.3.25}$$

Equations (3.3.12i) and (3.3.14) yield

$$u_z(r, \theta, z) = u_z(r, \theta, z_0) + \int_{z_0}^{z} \beta(z)\, dz. \tag{3.3.26}$$

Expressions (3.3.24) and (3.3.26) together with (3.3.8) lead to

$$\alpha(z) = c_1 z + c_2 \tag{3.3.27}$$ *

and

$$u_z(r, \theta, z_0) = u_z(r_0, \theta, z_0) + (r - r_0)\frac{\partial u_z}{\partial r}(r_0, \theta, z_0) + \frac{c_1}{6}(r - r_0)^2. \tag{3.3.28}$$

With (3.3.7), (3.3.27) implies that

$$\beta(z) = -2c_1 z + c_3.$$

Condition (3.3.15a) requires that

$$A(z) = c_4 z + c_5, \quad B(z) = c_6 z + c_7,$$

and application of (3.3.15b) gives

$$\mu(z) = c_8 z + c_9.$$

Gathering the results obtained to this point, we have

$$u_r(r, \theta, z) = (c_4 z + c_5)\cos\theta + (c_6 z + c_7)\sin\theta + r(c_1 z + c_2),$$
$$u_\theta(r, \theta, z) = -(c_4 z + c_5)\sin\theta + (c_6 z + c_7)\cos\theta + r(c_8 z + c_9),$$
$$u_z(r, \theta, z) = u_z(r, \theta, z_0) + c_3(z - z_0) - c_1(z^2 - z_0^2).$$

The three conditions (3.3.9), (3.3.10), and (3.3.11) have yet to be satisfied. Written in terms of the above form for the displacement, they are

$$-2c_1 r + \frac{\partial}{\partial r}\left(r\frac{\partial u_z}{\partial r}(r, \theta, z_0)\right) + \frac{1}{r}\frac{\partial^2 u_z}{\partial \theta^2}(r, \theta, z_0) = 0, \tag{3.3.29}$$

$$c_1 r - 3\left[\frac{\partial u_z}{\partial r}(r, \theta, z_0) - \frac{1}{r}\frac{\partial^2 u_z}{\partial \theta^2}(r, \theta, z_0)\right] = 0, \tag{3.3.30}$$

$$\frac{\partial}{\partial \theta}\left[\frac{1}{r}u_z(r, \theta, z_0) - \frac{\partial u_z}{\partial r}(r, \theta, z_0)\right] = 0. \tag{3.3.31}$$

By solving (3.3.29) for the second derivative of u_z with respect to θ and inserting the result into (3.3.30), we find that

$$\frac{\partial^2 u_z}{\partial r^2}(r, \theta, z_0) = \frac{5c_1}{3}.$$

Comparison of the solution of this equation,

$$u_z(r, \theta, z_0) = u_z(r_0, \theta, z_0) + (r - r_0)\frac{\partial u_z}{\partial r}(r_0, \theta, z_0) + \frac{5c_1}{6}(r - r_0)^2,$$

with (3.3.28) shows that

$$c_1 = 0,$$

so

$$u_z(r, \theta, z_0) = u_z(r_0, \theta, z_0) + (r - r_0)\frac{\partial u_z}{\partial r}(r_0, \theta, z_0). \tag{3.3.32}$$

* Here and in the remainder of this subsection c_i represents an arbitrary constant.

Substitution of (3.3.32) into (3.3.31) yields

$$\frac{\partial}{\partial\theta}\left[\frac{1}{r}u_z(r,\theta,z_0)-\frac{\partial u_z}{\partial r}(r,\theta,z_0)\right]=0,$$

which gives

$$u_z(r,\theta,z_0)=u_z(r,\theta_0,z_0)+r\left(\frac{\partial u_z}{\partial r}(r,\theta,z_0)-\frac{\partial u_z}{\partial r}(r,\theta_0,z_0)\right) \tag{3.3.33}$$

upon integration. Differentiation of (3.3.32) with respect to r results in

$$\frac{\partial u_z}{\partial r}(r,\theta,z_0)=\frac{\partial u_z}{\partial r}(r_0,\theta,z_0),$$

and evaluation of (3.3.32) at θ_0 yields

$$u_z(r,\theta_0,z_0)=u_z(r_0,\theta_0,z_0)+(r-r_0)\frac{\partial u_z}{\partial r}(r_0,\theta_0,z_0).$$

Incorporating these expressions into (3.3.33), one obtains

$$u_z(r,\theta,z_0)=rf(\theta)+c_{10}.$$

In order that (3.3.29) and (3.3.30) be satisfied, $f(\theta)$ must have the form

$$f(\theta)=c_{11}\cos\theta+c_{12}\sin\theta;$$

then (3.3.31) is automatically satisfied.

Thus, in the present context, *a displacement field is a universal solution if and only if it has the form*

$$\begin{aligned}\boldsymbol{u}(r,\theta,z)=&\left[(\bar{c}_1z+\bar{c}_2)\cos\theta+(\bar{c}_3z+\bar{c}_4)\sin\theta+\bar{c}_5r\right]\boldsymbol{e}_r\\&+\left[(\bar{c}_3z+\bar{c}_4)\cos\theta-(\bar{c}_1z+\bar{c}_2)\sin\theta+(\bar{c}_6z+\bar{c}_7)r\right]\boldsymbol{e}_\theta\\&+\left[(\bar{c}_8\cos\theta+\bar{c}_9\sin\theta)r+\bar{c}_{10}z+\bar{c}_{11}\right]\boldsymbol{e}_z,\end{aligned} \tag{3.3.34}$$

where the $\bar{c}_i$ are constants. Expression (3.3.34) contains the translation

$$(\bar{c}_2\cos\theta+\bar{c}_4\sin\theta)\boldsymbol{e}_r+(\bar{c}_4\cos\theta-\bar{c}_2\sin\theta)\boldsymbol{e}_\theta+\bar{c}_{11}\boldsymbol{e}_z,$$

which does not contribute to the stress. However, in contrast to the classical case, infinitesimal rotations, such as

$$\bar{c}_7r\boldsymbol{e}_\theta$$

in (3.3.34), do contribute to the stress through the $\boldsymbol{W}\mathring{\boldsymbol{T}}$ term in (2.3.1e).

Residual stress depends on r and z alone

In this case an analysis similar to the preceding with the residual stress represented by (3.2.12) leads to the conclusion that *a displacement field is a universal solution if and only if it has the form*

$$\begin{aligned}\boldsymbol{u}(r,\theta,z)=&\left[k_1\cos\theta+k_2\sin\theta+k_3r+(k_4\cos\theta+k_5\sin\theta)z\right]\boldsymbol{e}_r\\&+\left[k_2\cos\theta-k_1\sin\theta+k_6r+(k_5\cos\theta-k_4\sin\theta)z\right]\boldsymbol{e}_\theta\\&+\left[k_3z-(k_4\cos\theta-k_5\sin\theta)r+k_7\right]\boldsymbol{e}_z,\end{aligned} \tag{3.3.35}$$

where the k_i are constants. When the residual stress field possesses torsionless axisymmetry, the result is the same. Recognizing that

$$(k_1\cos\theta+k_2\sin\theta)\boldsymbol{e}_r+(k_2\cos\theta-k_1\sin\theta)\boldsymbol{e}_\theta+k_7\boldsymbol{e}_z$$

is a translation, we note that the displacement field (3.3.35) is even more limited than appears at first glance.

3.4. Determination of the incremental elastic constants

In this subsection (except in the last paragraph) and the next, it is presumed that the residual stress $\mathring{\boldsymbol{T}}$ depends only on r and θ and hence is given by (3.2.5); and, as throughout Section 3, $\mathbf{L}[\boldsymbol{E}] = 2\bar{\mu}\boldsymbol{E} + \bar{\lambda}(\text{tr } \boldsymbol{E})\mathbf{1}$. Universal solutions that are realizable in the laboratory are used in an inverse manner to determine the constitutive properties of the cylinder.

Consider the displacement field

$$\boldsymbol{u} = \eta r \boldsymbol{e}_r + \gamma z \boldsymbol{e}_z, \tag{3.4.1}$$

where η and γ are constants. Since this is a universal solution for the constitutive class at hand (cf. (3.3.34)), we know that it can be maintained in the cylinder by surface tractions alone no matter what $\bar{\mu}$, $\bar{\lambda}$, and $\mathring{\boldsymbol{T}}(r, \theta)$ may be. Obviously this is a crucial consideration in the experimental determination of these material properties.

The traction on the boundary corresponding to (3.4.1) is given by (2.3.7) as

$$\sigma = \begin{cases} [2\bar{\mu}\gamma + \bar{\lambda}(2\eta + \gamma)]\boldsymbol{e}_z \equiv \hat{\sigma}_z \boldsymbol{e}_z & \text{on } \partial B_z \\ [2\bar{\mu}\eta + \bar{\lambda}(2\eta + \gamma)]\boldsymbol{e}_r & \text{on } \partial B_r, \\ -\hat{\sigma}_z \boldsymbol{e}_z & \text{on } \partial B_0. \end{cases} \tag{3.4.2}$$

It is desirable to have the lateral surface ∂B_r traction free, and this can be arranged by taking

$$2\bar{\mu}\eta + \bar{\lambda}(2\eta + \gamma) = 0.$$

Coupling this with (3.4.2a), we find *the incremental elastic constants to be*

$$\bar{\mu} = \frac{-\hat{\sigma}_z}{2(\eta - \gamma)}, \quad \bar{\lambda} = \frac{\eta \hat{\sigma}_z}{(\eta - \gamma)(2\eta + \gamma)}. \tag{3.4.3}$$

The parameters η and γ are easily measured by noting from (3.4.1) that the length and diameter of the deformed cylinder are $(1 + \gamma)z_0$ and $2(1 + \eta)r_0$, respectively.

The situation described here can be physically realized by axially compressing the cylinder between lubricated rigid blocks. Since the applied traction $\hat{\sigma}_z$ is uniform, it is readily calculated from the load applied to the blocks.

Note that the residual stress does not appear in any explicit way in this boundary value problem, and the corresponding experiment is identical to the one used in the classical theory when $\mathring{\boldsymbol{T}} \equiv \boldsymbol{0}$. * This does not mean that the incremental elasticities are independent of the residual stress; it does mean that there is nothing in the tension test that can indicate either the presence or the absence of residual stress of the form $\mathring{\boldsymbol{T}}(r, \theta)$.

When $\mathring{\boldsymbol{T}}$ depends on r and z alone, it follows from (3.3.35) that (3.4.1) is a universal solution if and only if $\eta = \gamma$. Then it is clear from (3.4.3) that the tension test cannot be used to determine $\bar{\mu}$ and $\bar{\lambda}$ in this case. In fact, the strain corresponding to the most

* Actually, the customary interpretation in the classical theory is to put the cylinder in tension through loading grips and then invoke St. Venant's principle to conclude that (3.4.1) and (3.4.2) hold sufficiently far from the grips.

general universal solution (3.3.5) is $\boldsymbol{E} = k_3\mathbf{1}$, and by (2.3.7) and (3.1.8) the surface traction reduces to the uniform pressure

$$\sigma = (2\bar{\mu} + 3\bar{\lambda})k_3\boldsymbol{n}. \tag{3.4.4}$$

Thus one can determine the combination $2\bar{\mu} + 3\bar{\lambda}$ by subjecting the cylinder to a uniform pressure, but this is the most that can be learned by employing universal solutions in this case.

3.5 Determination of the residual stress field

The hypotheses of the previous subsection are adopted here, and in view of the results obtained there, we treat the incremental elastic constants $\bar{\mu}$ and $\bar{\lambda}$ as known.

Consider the universal solution *

$$\boldsymbol{u} = \alpha rz\boldsymbol{e}_\theta, \tag{3.5.1}$$

where α is a constant. By (2.3.7) the corresponding traction on the boundary is

$$\sigma = \begin{cases} -\dfrac{\alpha}{4}\left(\dfrac{1}{r}\phi_{,\theta} - \phi_{,r\theta}\right)\boldsymbol{e}_r - \dfrac{\alpha}{4}(r\theta_{,rr} - 4\bar{\mu}r)\boldsymbol{e}_\theta \equiv \hat{\sigma}_r(r,\theta)\boldsymbol{e}_r + \hat{\sigma}_\theta(r,\theta)\boldsymbol{e}_\theta & \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \text{on } \partial B_z, \\ \boldsymbol{0} \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \text{on } \partial B_r, \\ -\hat{\sigma}_r(r,\theta)\boldsymbol{e}_r - \hat{\sigma}_\theta(r,\theta)\boldsymbol{e}_\theta \qquad\qquad\qquad\qquad\qquad\qquad \text{on } \partial B_0. \end{cases} \tag{3.5.2}$$

Note that in this case the lateral surface ∂B_r is automatically free of traction. Since the stress function ϕ is independent of z, (3.5.2a) holds throughout the body, and we now solve this equation for ϕ.

Two integrations with respect to r of the θ component of (3.5.2a) yield

$$\phi(r,\theta) = \phi(r_0,\theta) + (r-r_0)\phi_{,r}(r_0,\theta) + 2\bar{\mu}(r-r_0)^2 - \frac{4}{\alpha}\int_{r_0}^{r}\int_{r_0}^{\rho}\frac{\hat{\sigma}_\theta(t,\theta)}{t}\,dt\,d\rho. \tag{3.5.3}$$

The expression

$$\phi_{,\theta}(r,\theta) = \phi_{,\theta}(r_0,\theta) + (r-r_0)\phi_{,r\theta}(r_0,\theta) - \frac{4}{\alpha}\frac{\partial}{\partial\theta}\int_{r_0}^{r}\int_{r_0}^{\rho}\frac{\hat{\sigma}_\theta(t,\theta)}{t}\,dt\,d\rho \tag{3.5.4}$$

follows from differentiation of (3.5.3) with respect to θ. Incorporation of the θ component of boundary condition (3.2.6) for the unloaded problem,

$$\phi_{,\theta}(r_0,\theta) - r_0\phi_{,r\theta}(r_0,\theta) = 0,$$

into (3.5.4) yields

$$\phi_{,\theta}(r,\theta) = r\phi_{,r\theta}(r_0,\theta) - \frac{4}{\alpha}\frac{\partial}{\partial\theta}\int_{r_0}^{r}\int_{r_0}^{\rho}\frac{\hat{\sigma}_\theta(t,\theta)}{t}\,dt\,d\rho. \tag{3.5.5}$$

Differentiation of (3.5.5) with respect to θ leads to

$$\phi_{,\theta\theta}(r,\theta) = r\phi_{,r\theta\theta}(r_0,\theta) - \frac{4}{\alpha}\frac{\partial^2}{\partial\theta^2}\int_{r_0}^{r}\int_{r_0}^{\rho}\frac{\hat{\sigma}_\theta(t,\theta)}{t}\,dt\,d\rho. \tag{3.5.6}$$

* Cf. the discussion following (3.4.1).

Substitution of (3.5.6) into the r component of boundary condition (3.2.6),

$$r_0\phi_{,r}(r_0, \theta) + \phi_{,\theta\theta}(r_0, \theta) = 0,$$

provides the following differential equation:

$$\phi_{,r\theta\theta}(r_0, \theta) + \phi_{,r}(r_0, \theta) = 0;$$

so

$$\phi_{,r}(r_0, \theta) = P \sin\theta + Q \cos\theta, \tag{3.5.7}$$

where P and Q are constants. Insertion of (3.5.7) into the θ component of (3.2.6) yields

$$\phi(r_0, \theta) = r_0(P \sin\theta + Q \cos\theta) + c, \tag{3.5.8}$$

where c is a constant. The stress function can now be written as

$$\phi(r, \theta) = 2\bar{\mu}(r - r_0)^2 - \frac{4}{\alpha}\int_{r_0}^{r}\int_{r_0}^{s}\frac{\hat{\sigma}_\theta(t, \theta)}{t}\,dt\,ds \tag{3.5.9}$$

as a consequence of (3.5.3), (3.5.7) and (3.5.8). We have dropped the additive term $r(P\sin\theta + Q\cos\theta) + c$ since it does not contribute to the stresses. It is easily confirmed that boundary condition (3.2.6) is satisfied.

The nonzero components of the residual stress are, from (3.2.5) and (3.5.9),

$$\left.\begin{aligned}
\mathring{T}_{rr}(r, \theta) &= \frac{4\bar{\mu}}{r}(r - r_0) - \frac{4}{\alpha r}\int_{r_0}^{r}\frac{\hat{\sigma}_\theta(t, \theta)}{t}\,dt - \frac{4}{\alpha r^2}\frac{\partial^2}{\partial\theta^2}\int_{r_0}^{r}\int_{r_0}^{\rho}\frac{\hat{\sigma}_\theta(t, \theta)}{t}\,dt\,d\rho,\\
\mathring{T}_{r\theta}(r, \theta) &= \frac{4}{\alpha r}\frac{\partial}{\partial\theta}\int_{r_0}^{r}\frac{\hat{\sigma}_\theta(t, \theta)}{t}\,dt - \frac{4}{\alpha r^2}\frac{\partial}{\partial\theta}\int_{r_0}^{r}\int_{r_0}^{\rho}\frac{\hat{\sigma}_\theta(t, \theta)}{t}\,dt\,d\rho,\\
\mathring{T}_{\theta\theta}(r, \theta) &= 4\bar{\mu} - \frac{4}{\alpha r}\hat{\sigma}_\theta(r, \theta).
\end{aligned}\right\} \tag{3.5.10}$$

Thus *the residual stress field is completely determined through (3.5.10) by the applied traction* $\hat{\sigma}_\theta$, *the angle of twist per unit length* α, *and the incremental elastic constant* $\bar{\mu}$.

Physically, the problem described here corresponds to twisting the cylinder between rigid blocks bonded to its ends. Since it follows from (3.5.1) that the relative angle of twist between the ends is αz_0, the parameter α is easily found. However, the pointwise measurement of the applied traction $\hat{\sigma}_\theta$ presents considerable experimental difficulties. Nonetheless, in contrast to the tension test, the torsion test * does have the potential for supplying information about the residual stress. In fact, lower bounds for the components of the residual stress in terms of the applied torque and α will be derived at the end of the present subsection.

It follows from (3.5.2a) and (3.5.9) that $\hat{\sigma}_r$ is given in terms of $\hat{\sigma}_\theta$ by

$$\hat{\sigma}_r(r, \theta) = -\frac{\partial}{\partial\theta}\int_{r_0}^{r}\frac{\hat{\sigma}_\theta(t, \theta)}{t}\,dt + \frac{1}{r}\frac{\partial}{\partial\theta}\int_{r_0}^{r}\int_{r_0}^{\rho}\frac{\hat{\sigma}_\theta(t, \theta)}{t}\,dt\,d\rho. \tag{3.5.11}$$

Consequently the residual stress component $\mathring{T}_{r\theta}$ may alternatively be expressed as

$$\mathring{T}_{r\theta}(r, \theta) = -\frac{4}{\alpha r}\hat{\sigma}_r(r, \theta). \tag{3.5.12}$$

* The usual interpretation in the classical theory is to twist the cylinder through grips rather than rigid blocks bonded to its ends (cf. the footnote in Subsection 3.4).

When the applied traction $\hat{\partial}_\theta$ is independent of θ, we see from (3.5.10) through (3.5.12) that

$$\mathring{T}_{r\theta} \equiv 0, \quad \hat{\partial}_r \equiv 0, \tag{3.5.13}$$

and $\mathring{\boldsymbol{T}}$ is a function of r alone with nonzero components

$$\left.\begin{aligned} \mathring{T}_{rr} &= \frac{4\bar{\mu}}{r}(r - r_0) - \frac{4}{\alpha r}\int_{r_0}^{r} \frac{\hat{\partial}_\theta(t)}{t}\,\mathrm{d}t, \\ \mathring{T}_{\theta\theta} &= 4\bar{\mu} - \frac{4}{\alpha r}\hat{\partial}_\theta(r). \end{aligned}\right\} \tag{3.5.14}$$

If, in addition, $\hat{\partial}_\theta$ is linear in r, i.e.,

$$\hat{\partial}_\theta(r, \theta) = \frac{\partial_0}{r_0} r \tag{3.5.15}$$

for some constant ∂_0, then (3.5.14) reduces to

$$\mathring{T}_{rr} = 4\left(\bar{\mu} - \frac{\partial_0}{\alpha r_0}\right)\frac{1}{r}(r - r_0), \quad \mathring{T}_{\theta\theta} = 4\left(\bar{\mu} - \frac{\partial_0}{\alpha r_0}\right). \tag{3.5.16}$$

Bounded behavior of $\mathring{T}_{rr}$ at the center of the cylinder requires

$$\partial_0 = \bar{\mu}\alpha r_0, \tag{3.5.17}$$

and then it follows from (3.5.13), (3.5.15) and (3.5.16) that

$$\mathring{\boldsymbol{T}} \equiv \boldsymbol{0} \tag{3.5.18}$$

and

$$\hat{\partial}_\theta(r, \theta) = \bar{\mu}\alpha r. \tag{3.5.19}$$

Conversely, suppose that (3.5.18) holds. Then (3.5.10c) implies (3.5.19). Thus *a necessary and sufficient condition for the absence of residual stress* (of the form $\mathring{\boldsymbol{T}}(r, \theta)$) *is that the applied shear stress $\hat{\partial}_\theta$ be independent of θ and linear in r.*

The z component of the resultant moment on the end ∂B_z is

$$T = \int_{\partial B_z} r\hat{\partial}_\theta(r, \theta)\,\mathrm{d}A. \tag{3.5.20}$$

Therefore by (3.5.2a)

$$T = \bar{\mu}J\alpha - \frac{\alpha}{4}\int_{\partial B_z} r^2\phi_{,rr}(r, \theta)\,\mathrm{d}A, \tag{3.5.21}$$

where

$$J = \frac{\pi r_0^4}{2} \tag{3.5.22}$$

is the polar moment of inertia of the cross section. Note that $\bar{\mu}J\alpha$ is the value of the applied torque in the classical case when $\mathring{\boldsymbol{T}} \equiv \boldsymbol{0}$. By (3.5.21) any deviation of the actual applied torque T from the classical value indicates the presence of a nonzero residual stress. In fact, we can get lower bounds on the components of the residual stress as follows.

By (3.2.5)

$$\phi_{,rr} = \mathring{T}_{\theta\theta},$$

and therefore (3.5.21) implies that

$$\max_B |\mathring{T}_{\theta\theta}| \geqslant \frac{2}{J|\alpha|}|T - \bar{\mu}J\alpha|. \tag{3.5.23}$$

We also can get a lower bound on $\mathring{T}_{rr}$ when $\mathring{T}$ depends on r alone. In this case $\phi = \phi(r)$, $\mathring{T}_{r\theta} \equiv 0$, and

$$\mathring{T}_{rr} = \frac{1}{r}\phi_{,r}, \quad \mathring{T}_{rr}(r_0) = 0 \tag{3.5.24}$$

(cf. (3.2.7), (3.2.8)). Integration by parts and substitution of (3.5.24) in (3.5.21) yield

$$T = \bar{\mu}J\alpha + \frac{3\pi}{2}\alpha\int_0^{r_0} r^3\mathring{T}_{rr}\,\mathrm{d}r.$$

Consequently

$$\max_B |\mathring{T}_{rr}| \geqslant \frac{1}{3J|\alpha|}|T - \bar{\mu}J\alpha|. \tag{3.5.25}$$

The bounds (3.5.23) and (3.5.25) make it clear that *if the applied torque T differs from the classical value $\bar{\mu}J\alpha$, then there is nonzero residual stress in the cylinder.* However, there may be residual stress present even when $T = \bar{\mu}J\alpha$.

References

[1] M.E. Gurtin, *An Introduction to Continuum Mechanics.* Academic Press, New York (1981).

[2] M.E. Gurtin, The linear theory of elasticity, *Handbuch der Physik,* vol. VIa/2, Springer-Verlag, Berlin (1972).

[3] Y.-H. Pao, W. Sachse and H. Fukuoka, Acoustoelasticity and ultrasonic measurements of residual stresses. *Physical Acoustics,* vol. XVII, W.P. Mason and R.N. Thurston eds., Academic Press, New York (1984).

[4] B.D. Coleman and W. Noll, Material symmetry and thermostatic inequalities in finite elastic deformations. *Arch. Rational Mech. Anal.* 15 (1964) 87–111.

[5] A. Hoger, On the residual stress possible in an elastic body with material symmetry. *Arch. Rational Mech. Anal.* 88 (1985) 271–290.

[6] G. Del Piero, Some properties of the set of fourth-order tensors, with application to elasticity. *J. Elasticity* 9 (1979) 245–261.

[7] Z.P. Bazant, A correlation study of formulations of incremental deformation and stability of continuous bodies. *J. Appl. Mech.* 38, (1971) 919–928.

[8] R.A. Toupin and B. Bernstein, Sound waves in deformed perfectly elastic materials. Acoustoelastic effect. *J. Acous. Soc. America* 33 (1961) 216–225.

[9] A.E. Green, Thermoelastic stresses in initially stressed bodies. *Proc. Roy. Soc. Lond.* A 266 (1962) 1–19.

[10] C. Truesdell and W. Noll, The non-liner field theories of mechanics. *Handbuch der Physik,* vol. III/3, Springer-Verlag, Berlin (1965).

[11] D.E. Carlson and C.-S. Man, Stability criteria sufficient for existence, uniqueness, and continuous dependence in linear elasticity with initial stress, in preparation.